

JORGE URDANETA

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EDUCATION

Bachelor of Science in Mechanical Engineering

The University of Texas at Austin

Expected Graduation: Dec 2026

Associate in Science in Mechanical Engineering, *Graduated with Honors*

Midland College

Graduated: 2023

TECHNICAL EXPERIENCE

Boeing Commercial Airplanes – Manufacturing Engineering Intern (777-777X Program)

June 2025 - August 2025

- Redesigned gyro battery housing in CATIA with integrated features and simplified wiring, cutting system footprint ~50%
- Engineered HUD inspection tool with GD&T to raise deviation sensitivity 75% and catch > .010" hard-point shifts
- Prototyped economy seat tray with collapsible cup holder and phone stand in 3-day design-a-thon, validating fit and usability
- Flagged 24 Boeing 777-9 aircraft with outdated pilot seat builds to drive rework efforts and align with program standards
- Verified cake stand alignment, created visual aid for installation approval, and supported aircraft LSM installation onsite

ASCEND Texas — GoAero & X-Prize Wildfire Challenge

Program Manager

May 2025 — Present

- Driving transition from ground prototype to autonomous hybrid aerial-terrestrial system, enabling remote payload delivery
- Built AppSheet coordination platform used by 40+ members, improving project tracking across 15+ individual projects
- Leading sponsor outreach across energy, aerospace, and emergency response sectors to secure funding and partnerships
- Secured GoAERO and NASA Innovation Grant Award for our ATV hybrid vehicle concept

Mechanical Design & Production Engineering Intern

Sept 2024 – May 2025

- Oversaw the removal of the gasoline engine and installation of electronic motor to convert ATV into fully electric vehicle
- Originated avionics enclosure concept prioritizing airflow, accessibility, and EMI shielding for system reliability
- Designed and led fabrication of suspended motor mount using C-channels, optimizing alignment and easing assembly
- Aligned drivetrain layout to meet spacing constraints for motor, batteries, and controller with upgrade potential

3D-Printed Robotic Arm – Robotics Engineer Personal Research

Present

- Modeled fully articulated robotic hand in CAD with smooth joint motion and tendon-driven actuation via fishing-line cables
- Transitioned actuator placement from palm to forearm to increase grip strength and load capacity
- Conducting research on load-cell, potentiometer, and EMG sensor integration for responsive feedback
- Designing components for 3D printing to enable low-cost replication for assistive-tech initiatives on campus

Mini-Bowling Mechanism – Robotics Engineer

Mar 2025

- Built and assembled a fully functional prototype including alley, gutters, pins, and a motor-actuated linkage mechanism
- Created a scoop attachment that mimics human wrist motion at the output joint of a four-bar linkage bowling mechanism
- Finalized manufacturing and assembly of components, ensuring alignment, tolerance control, and reliable performance

Mass Flow Meter Optimization System – Software Engineer

Jan 2023 – May 2023

- Implemented Python algorithms to improve measurement accuracy and efficiency, reducing error margins by over 10%
- Conducted root cause analysis on system inefficiencies, finding and implementing solutions to improve performance by 15%
- Collaborated with a cross-functional team to integrate process improvements and ensure reliable algorithm functionality

LEADERSHIP EXPERIENCE

Longhorn Energy Club – Vice-President of External Relations

Jun 2024 – May 2025

- Initiated partnerships with energy-related companies and organizations enhancing visibility and impact by +100%
- Showcased students' sustainable projects and innovative solutions to multiple companies in energy at our stellar events
- Organized panel series with HDR, FERVO, and Google experts on strategies for meeting data center energy demands

Society of Hispanic Professional Engineers (SHPE) – Social Media Co-Chair

Jun 2024 – May 2025

- Facilitated and participated in weekly tutoring and mentoring sessions for high school students pursuing STEM careers
- Led summer recruitment campaigns targeting incoming freshmen and transfer students, increasing SHPE's engagement and visibility through tailored social media content and outreach efforts and growing the organization by 100+ members
- Enhanced social media engagement strategies to increase community interaction by +565%

SKILLS & ACCOMPLISHMENTS

- Software:** Python, MATLAB, Finite Element Analysis (FEA), SolidWorks, OnShape, CATIA, Bambu Studio, ideaMaker
- Hardware:** CNC Machining, Manual Milling, Weldment, Machining Processes, Prototyping, Sensor Integration
- Awards:** 2nd place at Boeing BCA ONE Intern Design-a-thon, Hispanic Scholarship Fund Scholar, Nike Scholarship Finalist